

Question:

To understand and implement the basic programming constructs of Java, including variables, data types, operators, conditional statements, loops, and methods.

Steps

1. Import the `Scanner` class.
2. Create the `PartA` class.
3. Create the `studentDetails()` method to read student name, roll number, and marks.
4. Calculate total marks and percentage.
5. Create the `PartB` class.
6. Create methods to:
 - Check even or odd.
 - Find the largest of three numbers.
 - Find the grade using `if-else`.
 - Display the day using `switch`.
7. Create the `PartC` class.
8. Create methods to:
 - Print a multiplication table.
 - Display numbers from 1 to N.
 - Calculate the sum of first N natural numbers.
 - Generate the Fibonacci series.
9. Create the `PartD` class.

10. Create methods to:
 - Calculate factorial.
 - Check whether a number is prime.
 - Find the maximum of two numbers.
 - Calculate the area of a circle.
11. Create the `StudentUtilityProgram` class containing the `main()` method.
12. Create a `Scanner` object for user input.
13. Display the student utility menu.
14. Read the user's choice.
15. Use a `switch` statement to execute the selected operation.
16. Call the appropriate method from `PartA`, `PartB`, `PartC`, or `PartD`.
17. Display the result.
18. Repeat the menu using a `do-while` loop.
19. Close the `Scanner` when the program ends.
20. End the program.

Algorithm

1. **Start**
2. Create a `Scanner` object.
3. Display the Student Utility Program menu.
4. Read the user's choice.

5. If choice is **1**, read student details, calculate total marks and percentage, and display the result.
6. If choice is **2**, read a number and check whether it is even or odd.
7. If choice is **3**, read three numbers and find the largest.
8. If choice is **4**, read the percentage and determine the grade using **if-else**.
9. If choice is **5**, read a number from 1 to 7 and display the corresponding day using **switch**.
10. If choice is **6**, read a number and display its multiplication table from 1 to 10.
11. If choice is **7**, read **N** and display numbers from 1 to **N** using a **while** loop.
12. If choice is **8**, read **N** and calculate the sum of the first **N** natural numbers using a **for** loop.
13. If choice is **9**, read the number of terms and generate the Fibonacci series.
14. If choice is **10**, read a number and calculate its factorial.
15. If choice is **11**, read a number and check whether it is prime.
16. If choice is **12**, read two numbers and find the maximum.
17. If choice is **13**, read the radius and calculate the area of the circle.
18. If choice is **14**, display "**Program Ended**".

19. For any other choice, display "Invalid Choice".
20. Repeat the menu until the user chooses to exit.
21. Close the Scanner.
22. **Stop.**

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PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\venik\Documents\Records\week7> & 'C:\Program Fi
'-cp' 'C:\Users\venik\AppData\Roaming\Code\User\workspaceSto
\bin' 'StudentUtilityProgram'

===== STUDENT UTILITY PROGRAM =====
1. Student Information and Result
2. Check Even or Odd
3. Find Largest of Three Numbers
4. Display Grade
5. Display Day of the Week
6. Print Multiplication Table
7. Display Numbers from 1 to N
8. Calculate Sum of First N Natural Numbers
9. Generate Fibonacci Series
10. Find Factorial
11. Check Prime Number
12. Find Maximum of Two Numbers
13. Calculate Area of a Circle
14. Exit

Enter your choice: 11
Enter a number: 5
5 is a Prime Number
```